FOR A FLAT LOCALIZED ETALE, UNRAMIFIED, QUASI-FINITE, INTEGRAL OR GENERIC-FIBRE-UNIQUE MAP OF DOMAINS THE CONTRACTION OF ANY NONZERO IDEAL IS NONZERO

THE STACKS PROJECT AUTHORS

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Editorial note. Seven enumerated hypotheses appear in the counted statement span and the proof treats them in the order case (7) first and then the reduction from the others; that order is the author's and is reproduced. The seven references inside the counted proof span are recorded as non-load-bearing navigation pointers; they include cross-chapter tags and same-chapter label lines, and the applications that consume them stay inside the counted proof span.

Allowed standard prerequisites (editorial).

- Fibres of Spec(B) -> Spec(A) and the generic point, and the behaviour of the generic point under a flat injective map. used at proof lines 31723-31729 and 31743-31748. Is the geometry that lets case (7) be identified as the weakest hypothesis.
- Flat local homomorphisms are faithfully flat, so a flat map of domains is injective. used at proof lines 31740-31742. Guarantees that the generic point of Spec(B) maps to the generic point of Spec(A).
- The tensor product with the fraction field as a filtered colimit of localizations. used at proof lines 31733-31734. Converts the unit statement over B tensor_A K into a unit statement over some B_f, which is what produces the nonzero element of A cap J.
- No nontrivial specializations between fibre points for integral, quasi-finite, unramified and etale maps. used at proof lines 31749-31759. Reduces the remaining hypotheses to case (7).

Source reference. Original lemma 00PL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 935–935. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

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Source reference. Original lemma 00U5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1099–1099. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00GT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 619–619. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00HR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 653–653. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0CAN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15443–15443. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0B7C: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14100–14100. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02UR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3742–3742. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let A and B be domains and let $A \to B$ be a ring map. Assume $A \to B$ has additionally at least one of the following properties*

1. *it is the localization of an étale ring map,*
2. *it is flat and the localization of an unramified ring map,*
3. *it is flat and the localization of a quasi-finite ring map,*
4. *it is flat and the localization of an integral ring map,*
5. *it is flat and there are no nontrivial specializations between points of fibres of* $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$*,*
6. $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ *maps the generic point to the generic point and there are no nontrivial specializations between points of fibres, or*
7. *exactly one point of* $\mathrm{Spec}(B)$ *is mapped to the generic point of* $\mathrm{Spec}(A)$*.*

Then $A \cap J$ is nonzero for every nonzero ideal J of B.

Proof. Proof in case (7). Let K, resp. L be the fraction field of A, resp. B. By Algebra, Lemma 0CAN we see that the unique point of $\mathrm{Spec}(B)$ which maps to the generic point $(0) \in \mathrm{Spec}(A)$ is $(0) \in \mathrm{Spec}(B)$. We conclude that $B \otimes_A K$ is a ring with a unique prime ideal whose residue field is L (in fact it is equal to L but we do not need this). Choose $b \in J$ nonzero. Then b maps to a unit of L. Hence b maps to a unit of $B \otimes_A K$ (Algebra, Lemma 0B7C). Since $B \otimes_A K = \mathrm{colim}_{f \in A \setminus \{0\}} B_f$ we see that b maps to a unit of B_f for some $f \in A$ nonzero. This means that $bb' = f^n$ for some $b' \in B$ and $n \geq 1$. Thus $f^n \in A \cap J$ as desired.

In the rest of the proof, we show that each of the other assumptions imply (7). Under assumptions (1) – (5), the ring map $A \to B$ is flat and hence $A \to B$ is injective (since flat local homomorphisms are faithfully flat by Algebra, Lemma 00HR). Hence the generic point of $\mathrm{Spec}(B)$ maps to the generic point of $\mathrm{Spec}(A)$. Now, if there are no nontrivial specializations between points of fibres of $\mathrm{Spec}(B) \to$

$\mathrm{Spec}(A)$, then of course this generic point of $\mathrm{Spec}(B)$ has to be the unique point mapping to the generic point of $\mathrm{Spec}(A)$. So (6) implies (7). Finally, to finish we show that in cases (1) – (5) there are no nontrivial specializations between the points of fibres of $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$. Namely, see Algebra, Lemma 00GT for the integral case, Algebra, Definition 00PL for the quasi-finite case, and use that unramified and étale ring maps are quasi-finite (Algebra, Lemmas 02UR and 00U5). □